HAMZA MAJID

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SUMMARY

AI/ML Engineer combining 2+ years of production software engineering in regulated banking environments with AI/ML research at TU Berlin and an upcoming Software Development Engineer internship at Amazon Berlin. Currently pursuing an M.Sc. in Artificial Intelligence at Brandenburg University of Technology. Recent work includes an explainable RL agent for live 5G network slice control (research paper in preparation), a comparative study of GNNs and Transformers for trajectory prediction (61.1% vs. 45.8% vs. 39.3% baseline), financial time-series forecasting, and RAG pipelines. Seeking thesis opportunities in AI engineering and AI in finance in Germany.

WORK EXPERIENCE

Software Development Engineer Intern | Amazon | Berlin, Germany| 11/2026 – 04/2027

- Internship focus: distributed systems, cloud-native microservices and GenAI tooling

Working Student | Technische Universität Berlin | Berlin, Germany | 05/2026 – 10/2026

- Designed and trained MXRL, an explainable DQN agent (PyTorch, ~21k parameters dual temporal attention heads, concept encoder, dueling Q-head) for live E2-based 5G slice control, replacing rule-based control with a learned policy.
- Trained 100+ models via Stable-Baselines3 DQN (2M steps, 16 parallel environments on H100); MXRL outperformed a black-box MLP baseline across 10 seeds: higher reward (+5.51 vs +4.91; 95% CI +0.33–+0.87) and fewer false alarms (15.3% vs 21.6%).
- Added concept supervision (Normal / Degrading / Mitigated / Recovering) with class-weighted loss: concept macro-F1 0.88; mapping agreement 1.00 across seeds vs. 0.23–0.25 unsupervised.
- Built an InfluxDB/Grafana explainability dashboard (Docker) exposing per-UE delay observations, predicted vs. ground-truth concepts, attention heatmaps, V(s)/A(s,a)/Q(s,a) decomposition, Q-gap, and first-promotion decision with strongest alternative.
- Deployed full private 5G SA O-RAN testbed (OpenAirInterface + Open5GS core + FlexRIC Near-RT RIC); built KPI pipeline (E2 → SQLite → InfluxDB → Grafana) and automated all testbed operations via Python/shell scripts.
- Contributing to a research paper in preparation; delivered 100 trained models, reproducible pipeline, and full results documentation to the research team.

Working Student | Technische Universität Berlin | Berlin, Germany | 08/2025 – 04/2026

- Optimized the Autonomous Reservoir Computing pipeline, increasing time-series prediction accuracy by 65%.
- Enhanced the PyReco open-source library (reservoir computing/ESN) used by AI/ML researchers globally.
- Developed an interactive neural network visualization tool, improving reservoir structure understanding for new users.
- Implemented a robust error-handling framework using validator and decorator patterns, improving code reliability and user feedback.

Relocation to Germany | 05/2025 – 07/2025

- Relocation and settling in Germany, alongside full-time M.Sc. studies

Software Engineer | Bank Al Habib | Karachi, Pakistan | 08/2022 – 04/2025 (incl. Trainee, 08/2022 – 03/2023)

- Developed and maintained 9+ data-driven microservices using Java and Quarkus on Linux for core banking operations.
- Improved system performance by reducing execution time by 40% through algorithm and code optimization.
- Implemented Apache ActiveMQ based event-driven architecture, reducing inter-service latency by 95%.
- Designed IBM Db2 schemas and reporting solutions for financial analytics and risk monitoring.
- Contributed to agile development and code reviews, ensuring high-quality, production-ready systems.

EDUCATION

Master of Science in Artificial Intelligence | 04/2025 – present

Brandenburg University of Technology, Cottbus

Bachelor of Science in Computer Science | 09/2018 – 05/2022

Institute of Business Management, Karachi

SKILLS

- **Programming Languages:** Java, Python, C, MATLAB, SQL
- **Frameworks & Technologies:** Quarkus, Apache ActiveMQ, REST APIs, Micro-services Architecture, Spark, Databricks, Grafana, FlexRIC, OpenAirInterface
- **Database & Storage:** MySQL, PostgreSQL, DB2, InfluxDB, Data Warehousing
- **Data Collection:** Web Scraping (Python)
- **AI & Machine Learning:** Supervised & Unsupervised Learning, Neural Networks, Graph Neural Networks, Transformers, Reservoir Computing, Scikit-learn, PyTorch, OpenCV, LLM, NLP, Reinforcement Learning, Explainable AI (XAI)
- **Messaging & Event-Driven Systems:** Apache ActiveMQ, JMS Queues, Transaction Management
- **Quantitative & Financial Analysis**: Time-Series Analysis, Statistical Modeling, Regression, Forecasting, Optimization, Financial Data Analysis, Risk Metrics, Backtesting, Numerical Methods
- **Version Control & DevOps:** Git, GitHub, Jira, CI/CD, Docker
- **Tools & IDEs:** IntelliJ, VS Code, Eclipse; APIs: Postman/Insomnia
- **Languages**: English-C1, Urdu-Native, German-B1
- **MS Office** (Excel, PowerPoint)

PROJECTS

Graph-Based Trajectory Prediction using GNNs and Transformers

- Built a road network graph from OpenStreetMap data (~2K nodes, ~5K edges) and generated movement trajectories via random walks.
- Implemented GraphSAGE for spatial relationships and a Transformer for sequential patterns, with full training pipelines.
- Transformer achieved 61.1% top-1 accuracy vs. 45.8% for GraphSAGE and a 39.3% random baseline; ablation study showed trajectory context, not model capacity drove the advantage, concentrated at complex intersections..

Financial Time-Series Forecasting (Stock Market Prediction)

- Built a multi-ticker prediction pipeline for SPY, QQQ, and equities, comparing Logistic Regression, XGBoost, and LSTM using 55+ leakage-free technical and market-regime features.
- Implemented expanding-window walk-forward validation with purge/embargo gaps and majority-class baseline (AUC ~0.50–0.54), plus back testing with transaction costs, Sharpe, drawdown, and position-sizing overlays.
- Developed an interactive Streamlit dashboard for per-ticker training, walk-forward evaluation, back testing, and predictions, with optional LSTM support and transparent performance reporting.

Recall — Multi-Document RAG Pipeline

- Built a multi-document RAG pipeline with vector database retrieval, comparing chunking strategies and embedding models on retrieval quality. Evaluated with recall @ k metrics against a held-out question set.